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Generative artificial intelligence-mediated counselling on first aid for seizures: The performance of publicly available chatbot versus its customised version

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ABSTRACT

Background: The potential application of cutting-edge generative artificial intelligence chatbots in the capacity of emergency consultants is gaining growing attention. This study aimed to analyse the quality of advice on first aid for seizures generated by a commercially developed chatbot in comparison with its customised version.
Methods: The baseline version of ChatGPT (model GPT-4o) and the same chatbot customised using a specialised knowledge base and prompt engineering were tested in four scenarios mimicking bystander requests for instructions on how to help a victim with seizures. The scenarios included ongoing seizures and postictal states, with or without consciousness and breathing. A checklist-based evaluation was conducted.
Results: In total, 120 user-to-chatbot dialogues were generated (2 chatbots × 15 dialogues × 4 scenarios). The baseline chatbot always failed to consider the victim's state, including whether the seizures are continuing, or if the victim in the postictal period is conscious and breathing normally. Its advice was non-selective and inaccurate, with frequent omissions of key recommendations on first aid and suggestions of inadequate measures. The customised chatbot-generated guidance was consistently tailored to the victim's condition, significantly more precise and completely safe. Depending on the scenario, the mean percentage of chatbot responses that fulfilled the checklist items was 14–49 % for the baseline chatbot and 77–92 % for the customised version ($p \leq 0.039$).
Conclusions: Whereas the publicly available version of the chatbot is not acceptable for first aid counselling, its expert-informed customisation ensures high accuracy and safety of generated advice. Further research in this field is advisable.

1. Introduction

Generalised seizures are commonly occurring time-sensitive emergencies [1] that carry a significant risk of complications, including severe injuries [2], potentially fatal and disabling status epilepticus [3], and sudden unexpected death [4]. Prompt and correct application of first aid by a witness of generalised seizures may reduce the threat to the affected person's health and life and is therefore strongly encouraged [5].

Notwithstanding the importance of the immediate bystander response, public knowledge and skills of first aid for seizures are generally inadequate [6,7]. Insufficient competencies may prevent laypeople from attempting first aid or induce potentially harmful actions, which are supported by common misconceptions [5]. Along with proactive training on first aid, just-in-time provision of instructions on first aid to untrained incident witnesses by dispatchers of emergency medical services (EMS) over the telephone, also known as dispatch life support [8], represents a reasonable strategy to enhance community engagement in responding to health crises, including seizures. Although the public may have high expectations of receiving pre-arrival instructions on life-saving help from dispatchers [9], the availability of this resource-demanding counselling service is generally limited [10].

The recent emergence and progressive market penetration of generative artificial intelligence (GenAI) chatbots are offering new opportunities for immediate informational support of laypeople in emergencies. Trained on massive corpora of text data, these advanced

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conversational agents promptly comprehend user input and generate contextually appropriate responses, supporting dynamic and human-like dialogue [11]. Considering the rising popularity of GenAI and the anticipated multifold growth in the number of GenAI-capable smartphones in the coming years [12], the experimental evaluation of the chatbots' performance as pocket emergency consultants constitutes an important area of research.

Some studies have tested the ability of commercially developed GenAI chatbots to synthesise instructions on first aid for different health emergencies, presenting controversial conclusions concerning the accuracy and safety of the produced advice [13–20]. Two publications reported positive impressions of the chatbot-generated instructions on first aid for seizures, based on isolated observations [15,16]. However, there have been no attempts to conduct a comprehensive, structured evaluation of the performance of publicly available GenAI chatbots when instructing a layperson on how to provide first aid for seizures, and no research has investigated possibilities for enhancing the quality of GenAI-powered support through its expert-informed customisation.

This study aimed to perform a comparative analysis of the quality of advice on first aid for seizures generated by a commercially developed GenAI chatbot and an author-customised version of the chatbot. The authors supposed that the obtained results might inform future efforts to create effective and safe GenAI systems for counselling the general public on first aid, thereby contributing to improved health outcomes for people with seizures and other emergencies.

2. Material and methods

The study was conducted from March to May 2025. It involved the development and experimental testing of a GenAI chatbot tailored to

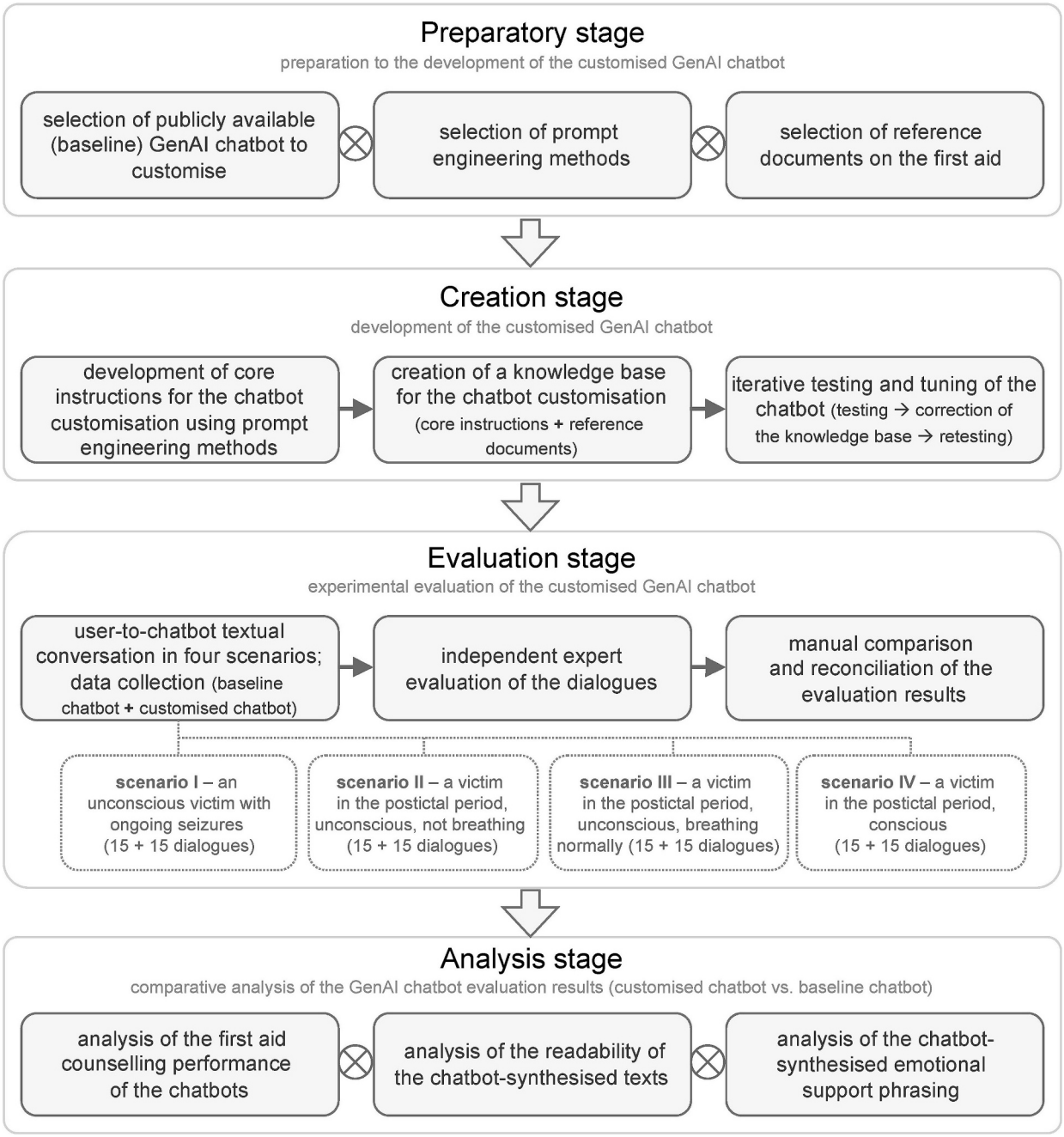


Fig. 1. Study flowchart.

provide emergency guidance on first aid for generalised seizures. The flowchart of the study is provided in Fig. 1.

The chatbot was developed as a custom version of publicly available ChatGPT (OpenAI, San Francisco, California, USA), model GPT-4o, using OpenAI's GPTs Editor web interface [21]. The following prompt engineering methods [22,23] were applied to attain optimal performance of the customised chatbot in terms of appropriateness, completeness, accuracy, clarity, and conciseness of the GenAI-mediated informational support: system prompting (creating a system block that specifies the overall objective, use cases and limitations); role prompting (assigning a specific role to the artificial intelligence model); contextual prompting (providing detailed context by listing the contents of the knowledge base); few-shot prompting (including command examples and response formats to demonstrate the required style); constraint setting (imposing strict limitations on permissible actions and topics); control of the max token length (limiting each message to 200 characters); iterative refinement and validation (multiple testing and refinement of prompts based on model error analysis and expert feedback); prompt templates (structuring prompts into logical blocks using templates); and tone prompting (explicitly defining the response style and tone).

The knowledge base of the customised chatbot (available in the Dataset [24]) comprised author-developed core instructions files and several reference files, including the International Federation of Red Cross (IFRC) and Red Crescent Societies' International First Aid, Resuscitation, and Education Guidelines 2020 (sections on first aid for seizures, unresponsiveness, and cardiac arrest) [25] and a selection of openly-available training materials and pre-arrival instructions for dispatcher-assisted cardiopulmonary resuscitation (CPR) [26–28]. The IFRC Guidelines were chosen as a primary source due to their authoritative status and widespread recognition as a global reference on evidence-based first aid practices. The educational materials, containing explanations of the principles and methods of dispatcher-led cardiac arrest identification and CPR coaching, and scripted pre-arrival instructions on CPR, were retrieved via Google using the search prompts "telephone CPR program" and "telephone CPR training resources". The core instructions files consisted of the following components: definitions of key terms and roles; communication and dialogue style guidelines; cataloguing and description of the knowledge base documents; operational recommendations for applying the knowledge base documents in a dialogue; prohibited actions; barrier mitigation strategies; chatbot phrasing examples; conversation closure instructions; and four sets of mandatory questions and the guidelines-based [25] instructional wordings corresponding to the four tested emergency scenarios described below.

Before the experimental evaluation, the operation of the customised chatbot was iteratively tested, and corrections were made to the knowledge base as appropriate to improve the performance. The working prototype of the customised chatbot is accessible online [29].

The experimental evaluation was carried out in four scenarios representing possible conditions of a victim who was found experiencing generalised seizures: scenario I – an unconscious victim with ongoing seizures; scenario II – a victim in the postictal period, unconscious, not breathing; scenario III – a victim in the postictal period, unconscious, breathing normally; scenario IV – a victim in the postictal period, conscious.

Data were generated repeatedly for each scenario through the user-to-chatbot textual conversation initiated by the authors with the following prompt: "Hello. I'm in the park, and there's a lady lying on the ground who seems to be having a seizure. I'm really worried about her. What can I do to help her until someone gets here?". When the chatbot performed the primary assessment of the victim's condition, inquiring whether the seizures were continuing, the victim was alert, or breathing normally, user responses were provided following the scenario using standardised phrasing (Appendix A). Additionally, for scenarios I-III, in response to the chatbot's first instruction on first aid implying the bystander's direct contact with the victim, bystander panic was introduced (see Appendix A) to test the ability of the chatbot to address the emotional distress of the bystander and ensure their adherence to the provided instructions.

Fifteen dialogues per scenario were synthesised and saved in text files for subsequent evaluation. The same number of dialogues was collected via a similar experiment with the publicly available ChatGPT, model GPT-4o (baseline chatbot) [30].

To assess the performance of the chatbots, a dialogue analysis was carried out using author-developed checklists informed by the international first aid guidelines [25] (see Dataset [24]). Two authors independently evaluated the dialogues, followed by a reconciliation of the obtained data.

The readability of the chatbot-synthesised texts was evaluated based on the Flesch-Kincaid Grade Level [31] calculated using the Datayze readability analyser [32]. The Flesch-Kincaid Grade Level measures the complexity of written content based on sentence length and word difficulty (syllable count). The resulting score corresponds to a U.S. education grade level, indicating the minimum level required to comprehend the text.

The analysis of emotional support phrasing generated by the chatbots in response to bystander panic, including the calculation of word counts and synthesis of word clouds, was carried out using Voyant Tools software [33].

The results were analysed using descriptive statistics and analytical statistics (Wilcoxon signed-rank test and Mann–Whitney *U* test) in IBM SPSS Statistics version 23.0 (IBM Corporation, Armonk, New York, USA).

3. Results

Both the baseline chatbot and the customised chatbot entered into a dialogue in all iterations for all scenarios. A total of 120 dialogues were collected for the analysis (2 chatbots × 15 dialogues × 4 scenarios). Examples of the dialogues are provided in Appendix B.

According to the readability evaluation, both chatbots generated simple texts that were easy to read and understand even for young readers or individuals with limited literacy skills. Compared to the baseline chatbot that created texts comprehensible to 8–10-year-olds (education grades 3–5), the customised chatbot offered higher readability (shorter sentences and fewer syllables per word) that corresponded to the level of kindergarten or the first year of elementary school (5–6 years of age; Table 1).

The baseline chatbot presented instructions on first aid combined in a block, with or without explicit headings indicating the key components of bystander intervention (see Appendix B). The customised chatbot, as trained, provided instructions sequentially, one at a time. Along with a generated instruction, this chatbot consistently asked the user for confirmation that the instruction had been followed and moved to the next instruction after receiving the confirmation.

Based on the statistical comparisons of checklist item coverage, the customised chatbot significantly outperformed the baseline chatbot in all scenarios (Tables 2–5). Across the scenarios, at the beginning of the conversation, the baseline chatbot never asked whether the victim was experiencing seizures. When the scenarios assumed the seizures had ended (scenarios II-IV), it never checked whether the victim was

Table 1
Readability of chatbot-generated text according to the Flesch-Kincaid Grade Level.

Scenario	Baseline chatbot ($N = 15$), median (interquartile range)	Customised chatbot ($N = 15$), median (interquartile range)	*p*-value
I	2.92 (2.79–3.54)	0.65 (0.53–0.76)	<0.001
II	3.15 (2.92–3.67)	0.42 (0.26–0.74)	<0.001
III	3.16 (3.06–3.31)	0.59 (0.37–0.86)	<0.001
IV	4.54 (4.16–4.59)	1.10 (0.86–1.50)	<0.001

Table 2
Performance of the chatbots in the scenario with the victim experiencing continued seizures (scenario I).

Checklist items	Chatbot responses that satisfied the checklist items, % (*n*)	
	Baseline chatbot (*N* = 15)	Customised chatbot (*N* = 15)
Primary assessment of the victim's condition		
1. The chatbot asks whether the victim is still experiencing seizures	0.0 (0)	100.0 (15)
Instructions on first aid		
2. The chatbot instructs the bystander to place soft padding under the victim's head to protect it	100.0 (15)	100.0 (15)
3. The chatbot instructs the bystander to move away any nearby objects that may hurt the victim	93.3 (14)	100.0 (15)
4. The chatbot instructs the bystander not to restrain the victim	100.0 (15)	100.0 (15)
5. The chatbot instructs the bystander to avoid moving the victim unless they are in immediate danger	13.3 (2)	6.7 (1)
6. The chatbot instructs the bystander to avoid trying opening the victim's mouth or placing any object in the victim's mouth	100.0 (15)	100.0 (15)
7. The chatbot instructs the bystander to remove victim's eyeglasses, if any	0.0 (0)	100.0 (15)
8. The chatbot instructs the bystander to loosen restrictive clothing from around the victim's neck, if any	33.3 (5)	100.0 (15)
9. The chatbot instructs the bystander to stay with the victim, monitor their condition, and get back once the seizure has ended	0.0 (0)	100.0 (15)
10. The chatbot instructs the bystander to note the start and stop time of the seizure, as well as if it reoccurs	100.0 (15)	60.0 (9)
Miscellaneous		
11. The chatbot verifies, after giving an instruction, whether the bystander has implemented it	0.0 (0)	always – 60.0 (9), in most instances – 40.0 (6)
12. The chatbot effectively addresses bystander emotional distress and hesitation, ensuring adherence to the instructions	100.0 (15)	100.0 (15)
13. The chatbot does not instruct the bystander to perform any superfluous actions	0.0 (0)	100.0 (15)
MEDIAN PERCENTAGE (interquartile range)	33.3 (0.0–100.0)	100.0 (80.0–100.0)*
MEAN PERCENTAGE (standard deviation)	49.2 (48.7)	86.7 (28.3)*

Note. * Significant difference ($p = 0.027$) between the baseline chatbot and the customised chatbot.

Table 3
Performance of the chatbots in the scenario with the victim in postictal period, unconscious and not breathing (scenario II).

Checklist items	Chatbot responses that satisfied the checklist items, % (*n*)	
	Baseline chatbot (*N* = 15)	Customised chatbot (*N* = 15)
Primary assessment of the victim's condition		
1. The chatbot asks whether the victim is still experiencing seizures	0.0 (0)	100.0 (15)
2. The chatbot asks whether the victim is alert	0.0 (0)	100.0 (15)
3. The chatbot asks whether the victim is breathing normally	13.3 (2)	100.0 (15)
Instructions on first aid		
4. The chatbot instructs the bystander to access emergency medical services	86.7 (13)	100.0 (15)
5. The chatbot instructs the bystander to get the victim flat on their back on the ground	0.0 (0)	100.0 (15)
6. The chatbot instructs the bystander to kneel by the victim's side	0.0 (0)	100.0 (15)
7. The chatbot instructs the bystander to put the heel of one hand on the centre of the victim's chest and put the other hand on top of the first hand	0.0 (0)	100.0 (15)
8. The chatbot instructs the bystander to push hard, at least 2 in. (5 cm)	0.0 (0)	100.0 (15)
9. The chatbot instructs the bystander to push fast, 100–120 times per minute	0.0 (0)	100.0 (15)
10. The chatbot instructs the bystander to push only on the heels of their hands	0.0 (0)	100.0 (15)
11. The chatbot instructs the bystander to keep their arms straight	0.0 (0)	100.0 (15)
12. The chatbot instructs the bystander to keep pumping the chest until help arrives	6.7 (1)	33.3 (5)
Miscellaneous		
13. The chatbot verifies, after giving an instruction, whether the bystander has implemented it	0.0 (0)	always – 53.3 (8), in most instances – 46.7 (7)
14. The chatbot effectively addresses bystander emotional distress and hesitation, ensuring adherence to the instructions	100.0 (15)	100.0 (15)
15. The chatbot does not instruct the bystander to perform any superfluous actions	0.0 (0)	100.0 (15)
MEDIAN PERCENTAGE (interquartile range)	0.0 (0.0–6.7)	100.0 (100.0–100.0)*
MEAN PERCENTAGE (standard deviation)	13.8 (32.6)	92.4 (20.3)*

Note. * Significant difference ($p = 0.001$) between the baseline chatbot and the customised chatbot.

conscious and, if unconscious (scenarios II and III), did not inquire whether the victim was breathing normally, except on two occasions for scenario II. As the baseline chatbot did not account for the victim's condition, the synthesised instructions had similar content throughout the scenarios. The advice typically included calling for EMS, protecting the victim from trauma, turning the victim onto their side, reassuring the victim, and timing the seizures. By contrast, the customised chatbot always performed the primary assessment of the victim's state and tailored the instructions on first aid to the recognised condition (see Tables 2–5).

In scenario I, both chatbots recommended protecting the convulsing victim from harm by putting soft padding under their head and moving away any surrounding objects they could hit, avoiding holding the victim down or putting anything in their mouth, but almost always left out the recommendation to avoid moving the victim unless they are in immediate danger (see Table 2). While the baseline chatbot showed better coverage of the instruction to time the seizure, it never advised removing the victim's eyeglasses and commonly omitted the instruction to loosen restrictive clothing on the victim's neck, if any, while the customised chatbot always did. The baseline chatbot never recommended staying with the victim and observing their condition, whereas the customised chatbot advised this in all instances. In all iterations, the baseline chatbot provided some superfluous instructions (see Dataset [24]). Most times, it was a recommendation to call EMS in any case ($n = 14$, 93.3 %) and a potentially harmful instruction to turn the victim onto their side ($n = 11$, 73.3 %). The customised chatbot did not provide any excessive advice.

In scenario II, based on the primary assessment of the victim's condition, the customised chatbot consistently recognised the victim's state as cardiac arrest, recommended calling EMS, and provided an

Table 4
Performance of the chatbots in the scenario with the victim in postictal period, unconscious and breathing normally (scenario III).

Checklist items	Chatbot responses that satisfied the checklist items, % (*n*)	
	Baseline chatbot (*N* = 15)	Customised chatbot (*N* = 15)
Primary assessment of the victim's condition		
1. The chatbot asks whether the victim is still experiencing seizures	0.0 (0)	100.0 (15)
2. The chatbot asks whether the victim is alert	0.0 (0)	100.0 (15)
3. The chatbot asks whether the victim is breathing normally	0.0 (0)	100.0 (15)
Instructions on first aid		
4. The chatbot instructs the bystander to access emergency medical services	86.7 (13)	100.0 (15)
5. The chatbot instructs the bystander to roll the victim on their side	66.7 (10)	100.0 (15)
6. The chatbot instructs the bystander to tilt the victim's head back	0.0 (0)	100.0 (15)
7. The chatbot instructs the bystander to stay with the victim and monitor the victim's breathing until emergency medical services arrive	6.7 (1)	40.0 (6)
Miscellaneous		
8. The chatbot verifies, after giving an instruction, whether the bystander has implemented it	0.0 (0)	always – 0.0 (0), in most instances – 100.0 (15)
9. The chatbot effectively addresses bystander emotional distress and hesitation, ensuring adherence to the instructions	100.0 (15)	100.0 (15)
10. The chatbot does not instruct the bystander to perform any superfluous actions	0.0 (0)	100.0 (15)
MEDIAN PERCENTAGE (interquartile range)	0.0 (0.0–71.7)	100.0 (85.0–100.0)*
MEAN PERCENTAGE (standard deviation)	26.0 (41.2)	84.0 (35.0)*

Note. * Significant difference ($p = 0.010$) between the baseline chatbot and the customised chatbot.

exhaustive set of directions on chest compressions, except the instruction to continue compressions until the arrival of help that was overlooked in two-thirds of cases (see Table 3). The baseline chatbot usually advised to call EMS, but the recommendation to start CPR was given only once, and it did not include any explanation of the chest compression technique. Further, the baseline chatbot always provided some (two to nine) superfluous directions inapplicable to the scenario (e.g. to protect the victim from injury, roll the victim onto their side, reassure the victim; see Dataset [24]), which was never the case for the customised chatbot.

In scenario III, the customised chatbot showed absolute precision in recognising the victim's condition as unresponsive and normally breathing and recommending to call EMS, turn the victim onto their side, and tilt their head back (see Table 4). However, the instruction to stay with the victim and observe their breathing until the arrival of help was provided in its entirety in fewer than half of the cases. The baseline chatbot totally failed to perform the primary assessment of the victim's condition, occasionally omitted the instruction to call EMS, frequently left out the advice to put the victim onto their side, and never recommended tilting the victim's head back. The instruction to stay by the victim's side and monitor their breathing until help arrives was given by the baseline chatbot on one occasion. While the customised chatbot did not provide any excessive instructions, the baseline chatbot included two to eight superfluous recommendations in each dialogue (see Dataset [24]). Many of these excessive suggestions could be relevant to continuing seizures (e.g. to move dangerous objects away, protect the victim's head, or avoid restraining the victim), but would be unhelpful and time-consuming in the tested scenario.

Table 5
Performance of the chatbots in the scenario with the victim in postictal period, conscious (scenario IV).

Checklist items	Chatbot responses that satisfied the checklist items, % (*n*)	
	Baseline chatbot (*N* = 15)	Customised chatbot (*N* = 15)
Primary assessment of the victim's condition		
1. The chatbot asks whether the victim is still experiencing seizures	0.0 (0)	100.0 (15)
2. The chatbot asks whether the victim is alert	0.0 (0)	100.0 (15)
Instructions on first aid		
3. The chatbot instructs the bystander to reassure the victim	100.0 (15)	100.0 (15)
4. The chatbot instructs the bystander to help the victim resting in the most comfortable position	0.0 (0)	100.0 (15)
5. The chatbot instructs the bystander to stay with the victim and monitor the victim's condition until full recovery or help arrives	0.0 (0)	46.7 (7)
Miscellaneous		
6. The chatbot verifies, after giving an instruction, whether the bystander has implemented it	0.0 (0)	always – 0.0 (0), in most instances – 100.0 (15)
7. The chatbot does not instruct the bystander to perform any superfluous actions	20.0 (3)	93.3 (14)
MEDIAN PERCENTAGE (interquartile range)	0.0 (0.0–20.0)	100.0 (46.7–100.0)*
MEAN PERCENTAGE (standard deviation)	17.1 (37.3)	77.1 (39.2)*

Note. * Significant difference ($p = 0.039$) between the baseline chatbot and the customised chatbot.

In scenario IV, the baseline chatbot in none of the cases performed the primary assessment of the victim's condition, always instructed to reassure the victim, never recommended to help the victim rest in the most comfortable position or stay with the victim and monitor their condition until complete recovery or arrival of help (see Table 5). Except for the latter recommendation, which was not fully reproduced in about half of the dialogues, the customised chatbot was accurate at all points. Whereas the guidance generated by the baseline chatbot contained one to seven superfluous instructions in 80.0 % ($n = 12$) of the iterations (see Dataset [24]), the customised chatbot included the unnecessary statement once (6.7 %), suggesting that the user check the victim's breathing.

In three scenarios (I-III) that tested the ability of GenAI to provide psychological support to a bystander experiencing panic, both chatbots in all instances successfully recognised the emotional distress and attempted to calm the bystander down and set them to implement the instructions on first aid (see Tables 2–4). The chatbots utilised similar vocabulary to generate empathic and reassuring content, underscoring that the bystander is doing the right thing by staying with the victim, that the bystander is not alone and will receive necessary guidance, and proposing to proceed with the first aid in a stepwise manner (Fig. 2). The baseline chatbot generated longer passages of text, which commonly incorporated controlled breathing techniques.

4. Discussion

There is little published evidence on the performance of GenAI tools in synthesising guidance for laypeople concerning how to provide help to a person with generalised seizures. A literature search carried out in May 2025 found two studies reporting relevant original data.

J.E. Dahdah and colleagues (2023) have investigated the ability of ChatGPT, model GPT-3.5, to answer questions covering 30 common

Baseline chatbot output 45 texts, 4713 total words, 331 unique word forms	Customised chatbot output 45 texts, 1254 total words, 115 unique word forms
Word cloud	Word cloud
	
Top-15 terms (counts) you're (211) just (118) right (86) doing (81) step (60) let's (47) breath (47) help (44) don't (43) deep (42) slowly (38) need (38) helping (34) perfect (33) breathe (30)	Top-15 terms (counts) you're (52) doing (52) step (42) right (33) i'll (28) helping (25) guide (24) just (21) hurt (20) let's (18) won't (17) worse (13) i'm (13) breath (13) time (11)
Phrasing example **You're** not alone—I'm here with you. Take a **deep breath**. You are already **helping**. **Just** by being there, **you're doing** more than most. **Let's** go **step**-by-**step** together. You **don't** have to do everything perfectly—**just** enough to keep her safe until **help** arrives. Right now, do this: **Breathe**. In through your nose… out through your mouth. **You're doing** great. Keep breathing with me. […] You **don't need** to fix this—you **just need** to protect her until **help** gets there.	Phrasing example **You're doing** the **right** thing **just** by being there. You can't make it **worse** — but **doing** nothing might. **I'll guide** you, one **step** at a **time**. **You're** not alone. **Let's** help her together.

Fig. 2. Outputs of the automated analysis of emotional support phrasing synthesised by the chatbots in response to bystander panic.

health emergencies, including"What to do if someone collapses in front of me and starts shaking?" [15]. Two emergency medicine specialists independently evaluated the generated responses and classified them as "appropriate", "inappropriate", or "harmful". Additionally, the responses were categorised as "easy", "acceptable", or "incomprehensible", depending on the number of terms that were considered confusing for the general public. The response to the question concerning collapse with "shaking" was classified as appropriate and easy to comprehend. Based on 30 evaluations (one chatbot response per emergency), the authors have concluded that ChatGPT can work as a "reliable 'triage' assistant" for medical emergencies and recommended future regular updates of the chatbot to maintain its medical credibility [15].

H.W. Kim and co-authors (2024) have assessed the accuracy and reliability of ChatGPT, models GPT-3.5 and GPT-4, in responding to 57 common questions on epilepsy, including two questions concerning first aid for seizures ("What is the first aid treatment for seizures?" and "What should I do if my child has a seizure?") [16]. The responses synthesised by ChatGPT (one response per question per model) were evaluated and classified by two epileptologists as "sufficient educational value", "correct but inadequate", "mixed with correct/incorrect/outdated information", and "incorrect". Responses to both first aid-related questions received the highest rating, and the tested models showed similar performance. Along with highlighting the potential of ChatGPT as a tool for delivering reliable information on epilepsy, the researchers have noted the importance of exercising caution when relying exclusively on GenAI tools for acquiring medical information and recommended further research [16].

Based on the comprehensive methodology that employed structured quantitative evaluation of multiple ChatGPT responses generated for different hypothetical emergency scenarios, the current study provides further evidence and insight into the efficacy, safety and flexibility of the cutting-edge GenAI tool as a means of real-time counselling on first aid for generalised seizures.

Both the baseline chatbot and the customised chatbot responded to all user requests by generating highly readable content comprehensible to elementary-level students, with the customised version demonstrating superior readability. The chatbots always recognised and attempted to manage bystander panic and hesitation by providing targeted emotional support and reassurance. As for the first aid guidance, the baseline chatbot presented multiple instructions as a single block of

text and did not check whether particular directions were followed by a user. Supposedly, this may adversely impact the completeness and effectiveness of the implementation of the instructions in real-life settings. By contrast, the customised chatbot generated instructions one after another and regularly verified whether an instruction was implemented before moving on to the next one, which may enhance user compliance with the provided advice.

The results of the checklist-based evaluation showed that the publicly available, baseline version of ChatGPT is not acceptable for first aid counselling, mainly because of poor selectivity and accuracy of generated guidance. At the start of the dialogue, it never inquired whether the seizures were continuing at the moment and did not ask about the victim's responsiveness and breathing. Consequently, the provided first aid advice, being similar across the tested scenarios, was not targeted to a specific victim's condition. It represented a conglomeration of instructions, some of which were appropriate for a particular case, while others were inadequate. The shortcomings of the synthesised guidance extended from the omission of relevant evidence-based recommendations on first aid to the provision of excessive instructions, varying from irrelevant to potentially harmful. For instance, the baseline chatbot never attempted to recognise cardiac arrest based on the consciousness and breathing assessment after the seizures had ended and did not provide instructions on chest compressions when the victim was unresponsive and not breathing normally. This is a major defect of emergency support since seizures are tightly associated with cardiac arrest. On the one hand, generalised tonic-clonic seizures represent a well-known risk factor for unexpected cardiac arrest and death in epilepsy [34], and on the other, transient seizure-like activity is a common symptom of the onset of sudden cardiac arrest [35]. An example of harmful advice commonly generated by the baseline chatbot included a recommendation to put the victim with continuing seizures in a recovery position. Forcibly positioning a person onto their side when the convulsions are ongoing is associated with the risk of potentially disabling shoulder dislocations [36]. The mean percentage of baseline chatbot responses that fulfilled the checklist items varied depending on the scenario, but never exceeded 50 %, while the proportion of responses containing superfluous instructions was 80 % or higher.

These results corroborate the findings of previous experimental studies that have demonstrated poor quality of advice on CPR and first aid synthesised by publicly available GenAI tools, including generation of incomplete, inaccurate, wrong, nonsensical and dangerous instructions [13,14,17–19]. Considering the exceptional public interest and rapidly progressing penetration of GenAI technology into society, the issue of GenAI-mediated health misinformation necessitates immediate and careful attention. Alongside enhancing public awareness concerning the risks associated with reliance on GenAI-generated first aid advice [17], establishing close expert oversight of GenAI services providing first aid counselling [20] and forbidding unsupervised generation of GenAI guidance on first aid of questionable quality [13], the expert-led development of custom GenAI architectures tailored for evidence-based instructional support of laypeople in health emergencies [19] represents a reasonable forward-looking solution for mitigating misinformation spread while retaining the benefits offered by the cutting-edge communication technology.

The customisation of ChatGPT using the specialised knowledge base, comprising current guidelines on first aid along with the principles and protocols of pre-arrival dispatch life support, and the set of author-developed prompts and rules that determined the chatbot's behaviour as a conversational emergency assistant, ensured its diagnostic selectivity and high reliability of the generated guidance. The customised chatbot always performed a primary assessment of the victim's condition and provided comprehensive evidence-based advice on first aid in strict accordance with the reported state of the victim, depending on whether they had continuing seizures or were responsive and normally breathing in the postictal period. When compared to the baseline chatbot guidance, advice synthesised by the customised version had significantly better coverage of relevant instructions on first aid. Furthermore, with the one exception, the customised chatbot did not provide unnecessary instructions, and it never suggested potentially harmful actions.

Notwithstanding the significantly higher performance of the customised version, the accuracy of generated advice was not absolute. The shortcomings included the omission of the instruction to avoid moving the victim with continuing seizures unless they are in danger, to note the start and stop time of the seizure, to continue chest compressions for cardiac arrest victim until the arrival of EMS, and to stay with the victim after the seizures have stopped and monitor their condition until their full recovery or the help arrives. The imperfections suggest that before the customised context-aware GenAI assistants are recommended for real-life emergency counselling, they should be subjected to a series of iterative evaluations and targeted adjustments, and undergo rigorous validation in randomised controlled trials with simulated environments. It is reasonable to perform further testing with various mobile devices (e.g. smartphones or smartwatches) and using different modes of conversation, including voice-based communication. Alongside comprehensive preclinical testing, an institution of robust expert-informed quality control and assurance procedures is indispensable to guarantee the reliability and safety of GenAI support in first aid in everyday settings.

4.1. Limitations

This study has limitations to consider. One of them is related to the use of a particular GenAI model at a particular moment in time. The technology is rapidly evolving and GenAI tools, including chatbots, are receiving numerous developer updates that influence their performance. Hence, repeated evaluations with other GenAI chatbots or the same chatbot but at a later date may produce different results. Another limitation is related to the textual form of user-to-chatbot conversation instead of the voice interaction that could be more appropriate for time-constrained settings of health emergencies. Further, the conversations in the current research followed particular dialogue scripts, which defined scenario-specific and simplified lines of user behaviour. Consequently, the experiment did not account for potential confounders associated with the varied behaviours of different users that may influence the performance of the chatbots. Further research with human participants in simulated settings is therefore advisable. Finally, the evaluation of the chatbots' performance was conducted by the authors themselves, potentially introducing bias.

5. Conclusions

The current publicly available version of ChatGPT readily generates instructions on first aid for seizures on a user request. However, the baseline chatbot version fails to consider the victim's state at the time of conversation, including whether the seizures are continuing or if the victim in the postictal period is conscious and breathing normally. Its advice is, therefore, non-selective and inaccurate, with frequent omissions of key recommendations on first aid and suggestions of excessive measures, including potentially harmful actions. The expert-guided customisation of the chatbot resulted in the creation of a specialised, context-aware conversational assistant that consistently provided recommendations on first aid tailored to the victim's condition, significantly outperforming the baseline version in the completeness and accuracy of generated instructions while ensuring the safety of the advice.

CRediT authorship contribution statement

Alexei A. Birkun: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yekaterina Kosova:** Writing – review & editing, Writing –

original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anton Rudenko:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A and Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.yebeh.2025.110680.

Data availability

The dataset generated and analysed during the current study is available in the Mendeley Data repository [24].

References

[1] Kämppi L, Puolakka T, Ritvanen J, Tuppurainen K, Päkkilä J, Kuisma M, et al. Burden of suspected epileptic seizures on emergency services: a population-based study. Eur J Neurol 2023;30(8):2197–205.
[2] Mühlenfeld N, Störmann P, Marzi I, Rosenow F, Strzelczyk A, Verboket RD, et al. Seizure related injuries - frequent injury patterns, hospitalization and therapeutic aspects. Chin J Traumatol Zhonghua Chuang Shang Za Zhi 2022;25(5):272–6.
[3] Choi SA, Lee H, Kim K, Park SM, Moon HJ, Koo YS, et al. Mortality, disability, and prognostic factors of status epilepticus: a nationwide population-based retrospective cohort study. Neurology 2022;99(13):e1393–401.
[4] Holst AG, Winkel BG, Risgaard B, Nielsen JB, Rasmussen PV, Haunsø S, et al. Epilepsy and risk of death and sudden unexpected death in the young: a nationwide study. Epilepsia 2013;54(9):1613–20.
[5] Sun M, Meng F, Xu ZYR, Guo Y. Seizure first aid in the community: current situation, suggestions, and the role of the general practitioner in seizure management. Acta Epileptol 2025;7(1):11.
[6] Al-Dosary AS, AlGhamdi FM, Almutairi BF, Alquwaiz IAI, Alsomali AM, Algarni SA, et al. Public awareness of first-aid management of seizures in Saudi Arabia. Epilepsy Behav EB 2022;129:108634.
[7] Adal O, Abebe A. First aid knowledge and practice toward students with epileptic seizure among governmental high school teachers in Addis Ababa, Ethiopia: cross-sectional study. Epilepsy Behav EB 2022;134:108767.
[8] Yancey A, Clawson JJ. EMD position paper resource document. AEDR 2014;2(2):33–40.
[9] Billittier AJ, Lerner EB, Tucker W, Lee J. The lay public's expectations of prearrival instructions when dialing 9-1-1. Prehosp Emerg Care 2000;4(3):234–7.
[10] Zhang L, Luo M, Myklebust H, Pan C, Wang L, Zhou Z, et al. When dispatcher assistance is not saving lives: assessment of process compliance, barriers and outcomes in out-of-hospital cardiac arrest in a metropolitan city in China. Emerg Med J EMJ 2021;38(4):252–7.
[11] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. Adv Neural Inf Process Syst 2017;30.
[12] GenAI-capable Smartphone Shipments to Grow Over 4X by 2027, https://www.counterpointresearch.com/insights/genai-capable-smartphone-shipments-to-grow-over-4x-by-2027/; 2024 [accessed 01 July 2025].
[13] Birkun AA, Gautam A. Large language model (LLM)-powered chatbots fail to generate guideline-consistent content on resuscitation and may provide potentially harmful advice. Prehospital Disaster Med 2023;38(6):757–63.
[14] Bushuven S, Bentele M, Bentele S, Gerber B, Bansbach J, Ganter J, et al. "ChatGPT, can you help me save my child's life?" - diagnostic accuracy and supportive capabilities to lay rescuers by ChatGPT in prehospital basic life support and paediatric advanced life support cases – an in-silico analysis. J Med Syst 2023;47(1):123.
[15] Dahdah JE, Kassab J, Helou MCE, Gaballa A, Sayles S, Phelan MP. ChatGPT: a valuable tool for emergency medical assistance. Ann Emerg Med 2023;82(3):411–3.
[16] Kim HW, Shin DH, Kim J, Lee GH, Cho JW. Assessing the performance of ChatGPT's responses to questions related to epilepsy: a cross-sectional study on natural language processing and medical information retrieval. Seizure Eur J Epilepsy 2024;114:1–8.
[17] Knebel D, Priglinger S, Scherer N, Klaas J, Siedlecki J, Schworm B. Assessment of ChatGPT in the prehospital management of ophthalmological emergencies – an analysis of 10 fictional case vignettes. Klin Monatsblätter Für Augenheilkd 2024;241(05):675–81.
[18] Yau JYS, Saadat S, Hsu E, Murphy LSL, Roh JS, Suchard J, et al. Accuracy of prospective assessments of 4 large language model chatbot responses to patient questions about emergency care: experimental comparative study. J Med Internet Res 2024;26:e60291.
[19] Aqavil-Jahromi S, Eftekhari M, Akbari H, Aligholi-Zahraie M. Evaluation of correctness and reliability of GPT, Bard, and Bing chatbots' responses in basic life support scenarios. Sci Rep 2025;15(1):11429.
[20] Pandya S, Alessandri Bonetti M, Liu HY, Jeong T, Ziembicki JA, Egro FM. Burn patient education in the modern age: a comparative analysis of ChatGPT and Google performance answering common questions on burn injury and management. J Burn Care Res 2025:irae211.
[21] Creating a GPT. OpenAI Help Center, https://help.openai.com/en/articles/8554397-creating-a-gpt; 2025 [accessed 01 July 2025].
[22] Patil R, Heston TF, Bhuse V. Prompt engineering in healthcare. Electronics 2024;13(15):2961.
[23] Google. Whitepaper on Prompt Engineering by Google, http://archive.org/details/whitepaper-prompt-engineering-v-4; 2024 [accessed 01 July 2025].
[24] Birkun A, Kosova Y, Rudenko A. A dataset comparing the performance of a publicly available generative artificial intelligence chatbot and its customised version in providing first aid guidance for seizures. Mendeley Data 2025:V1. https://doi.org/10.17632/6h53jrhf7t.1.
[25] International Federation of Red Cross and Red Crescent Societies. International first aid, resuscitation and education guidelines 2020, https://www.ifrc.org/document/international-first-aid-resuscitation-and-education-guidelines; 2020 [accessed 01 July 2025].
[26] Medical College of Wisconsin. Telecommunicator CPR Program Dispatch Protocols In-House, https://www.mcw.edu/-/media/MCW/Departments/Emergency-Medicine/EMS/Telecommunicator-CPR-Program-Dispatch-Protocols-In-House.pdf; 2019 [accessed 01 July 2025].
[27] CPR LifeLinks: Implementation Toolkit, https://www.ems.gov/assets/CPR_LifeLinks_Toolkit_2023_Final.pdf; [accessed 01 July 2025].
[28] Resuscitation Academy. T-CPR Toolkit, https://www.resuscitationacademy.org/s/DACPRToolkit1010.pdf; 2017 [accessed 01 July 2025].
[29] ChatGPT. LIRA – Life-saving Intelligent Response Assistant, https://chatgpt.com/g/g-67d864a14a588191a3f5224661559d34-lira; 2025 [accessed 01 July 2025].
[30] ChatGPT, https://chatgpt.com; 2025 [accessed 01 July 2025].
[31] Kincaid J, Fishburne R, Rogers R, Chissom B. Derivation Of New Readability Formulas (Automated Readability Index, Fog Count And Flesch Reading Ease Formula) For Navy Enlisted Personnel. Inst Simul Train 1975 Jan 1.
[32] Datayze. Readability Analyzer, https://datayze.com/readability-analyzer; [accessed 01 July 2025].
[33] Voyant Tools, https://voyant-tools.org/; [accessed 01 July 2025].
[34] Boglietti E, Haddad D, Bezin L, Rheims S. Pathophysiology of SUDEP: how far are we from understanding? Rev Neurol (Paris) 2025;181(5):432–7.
[35] Murasaka K, Takada K, Yamashita A, Ushimoto T, Wato Y, Inaba H. Seizure-like activity at the onset of emergency medical service-witnessed out-of-hospital cardiac arrest: an observational study. Resusc Plus 2021;8:100168.
[36] DeToledo JC, Lowe MR. Seizures, lateral decubitus, aspiration, and shoulder dislocation: time to change the guidelines? Neurology 2001;56(3):290–1.